

1 Solution — GIAM 7.1

Problem 5

1.1 Problem

The “*Cryptographer’s alphabet*” is used to supply small examples in coding and cryptography. It consists of the first 6 letters, $\{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}, \mathbf{f}\}$. How many “words” of length up to 6 can be made with this alphabet? (A word need not actually be a word in English: for example both “**fed**” and “**dfe**” would be words in the sense we are using the term.)

1.2 Setup

A word of length ℓ over the alphabet $\Sigma = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}, \mathbf{f}\}$ is an ordered ℓ -tuple of letters from Σ , with repetition allowed. Two consequences flow from this:

- **Order matters.** “**fed**” and “**dfe**” are distinct length-3 words, exactly as the problem statement emphasizes.
- **Repetition is allowed.** “**aaa**” is a valid length-3 word, even though only one distinct letter appears.

We interpret “length up to 6” as $\ell \in \{1, 2, 3, 4, 5, 6\}$ — i.e., the empty word is excluded — and revisit the alternative reading in the remark.

1.3 Count

For each fixed length ℓ , the multiplication principle gives 6^ℓ words: each of the ℓ positions in the tuple is filled independently by one of the 6 letters of Σ .

Length ℓ	Number of words 6^ℓ
1	6
2	36
3	216

4	1,296
5	7,776
6	46,656

Table 1.1

Words of different lengths are *disjoint* — a single word has exactly one length — so the addition principle applies:

$$\#\text{words of length up to 6} = \sum_{\ell=1}^6 6^\ell = 6 + 36 + 216 + 1,296 + 7,776 + 46,656 = \boxed{55,986}.$$

1.4 Closed Form via Geometric Series

The sum is a finite geometric series with first term 6, common ratio 6, and 6 terms.

The standard formula $\sum_{\ell=1}^N r^\ell = r \cdot \frac{r^N - 1}{r - 1}$ gives

$$\sum_{\ell=1}^6 6^\ell = \frac{6 \cdot (6^6 - 1)}{6 - 1} = \frac{6 \cdot 46,655}{5} = \frac{279,930}{5} = 55,986.$$

Equivalently,

$$\sum_{\ell=1}^6 6^\ell = \frac{6^7 - 6}{5} = \frac{279,936 - 6}{5} = 55,986,$$

using $6^7 = 279,936$. Both forms agree with the term-by-term tally.

1.5 Remark — The Empty Word

If one prefers to include the empty word ε (the unique word of length 0) — common in formal language theory and many cryptographic contexts — the count increases by exactly 1:

$$\sum_{\ell=0}^6 6^\ell = 1 + 55,986 = 55,987 = \frac{6^7 - 1}{5}.$$

This is the standard $\sum_{\ell=0}^N r^\ell = \frac{r^{N+1} - 1}{r - 1}$ form of the geometric sum. The natural English reading of "*words that can be made*" — and the examples "**fed**", "**dfe**" given in the problem statement — point to nonempty words, so we report **55,986** as the primary answer.

1.6 A Different Decomposition (Cross-Check)

The total **55,986** can also be obtained without invoking the geometric series, by separating the longest length out:

$$\sum_{\ell=1}^6 6^\ell = 6^6 + \sum_{\ell=1}^5 6^\ell = 46,656 + 9,330 = 55,986,$$

where the sub-sum $\sum_{\ell=1}^5 6^\ell = 6 + 36 + 216 + 1,296 + 7,776 = 9,330$ counts all shorter words. The length-**6** stratum alone already accounts for over **83%** of all words of length up to **6** — a useful intuition that the count is dominated by the maximum length when the alphabet is bigger than **1**.